

Enterprise Async RAG KnowledgeBase — One Page

What it is

Enterprise-grade, privately deployable RAG platform built on Spring Boot 3.5 and Milvus V2, optimized for hybrid dense/sparse retrieval and streaming chat.

Who it's for

Platform / AI engineering teams that need a low-latency, on-prem RAG stack with controllable ingestion, retrieval, and traceable responses.

What it does (key features)

- Dual-path dense + sparse recall via TEI; server-side RRF fusion in Milvus reduces network hops.
- Local reranker (BGE) with circuit breaker for graceful degradation when latency spikes.
- End-to-end asynchronous ETL: Tika parsing, chunking, embedding, and writes run in parallel.
- Backpressure + rate limiting on ingestion to protect embedding services under high concurrency.
- SHA-256 deduplication for idempotent uploads and resource savings.
- Structured prompt assembly with metadata injection and max-context trimming to avoid overflow.
- SSE streaming responses for chat-style UX.

How it works (architecture)

- Backend: Spring Boot 3.5 + Reactor orchestrates chat and ETL pipelines.
- Vector DB: Milvus V2 stores dense & sparse vectors; performs server-side RRF.
- Embedding service: Python TEI endpoint (port 8098) generates dense and sparse vectors.
- Rerank service: TEI Cross-Encoder (port 8099) with circuit breaker for availability.
- Storage & cache: MySQL for metadata; Redis for session/stream caching.
- LLM generation: Ollama-hosted model (e.g., qwen2.5:3b) streams replies via SSE.
- Observability: Jaeger + OTEL collector in Docker for tracing/metrics.

How to run (minimal)

- 1 Prereqs: Docker and JDK 17+ installed; ensure ./models/bge-reranker-base exists for rerank-service.
- 2 Start infra: run `docker-compose up -d` from repo root to launch Milvus, MySQL, Redis, Jaeger, OTEL, rerank-service.
- 3 Start embedding service: `cd embedding-service && pip install -r requirements.txt && python server.py` (listens on 8098).
- 4 Pull LLM: `ollama pull qwen2.5:3b` (or preferred) and keep Ollama running on 11434.
- 5 Run backend: `cd backend && mvn spring-boot:run` (HTTP 8080, SSE streaming).

Source: README.md (2026-03-06). All items reflect repo evidence; mark as Not found in repo if unavailable (none noted).